Unit 2 Assignment 2

Matrix Algebra & Systems of Equations

Learning Objectives

- ✅ Represent real-world data using matrices.
- ✅ Perform matrix operations manually and using NumPy.
- ✅ Apply matrix multiplication in AI.
- ✅ Understand determinant, inverse, identity matrix, transpose, and rank.
- ✅ Solve systems of equations using NumPy.
- ✅ Connect Linear Algebra with Machine Learning.

Instructions

> **Please read carefully**

- Work as one team.
- Discuss every solution.
- Show your manual calculations.
- Verify every numerical answer using NumPy.
- Submit **one notebook** and **one report** per group.

Dataset

House	Area	Bedrooms	Age
H1	120	3	10
H2	150	4	8

House	Area	Bedrooms	Age
H3	180	4	6
H4	210	5	4

Mission 1 — Understanding the Matrix

Tasks

☐ Represent the dataset as a matrix.

☐ Determine the matrix shape.

☐ Determine:

- Number of rows
- Number of columns

☐ Explain:

- What does each row represent?
- What does each column represent?

☐ Create the matrix using NumPy.

> **AI Insight**

What do rows and columns represent in Machine Learning?

Mission 2 — Matrix Operations

Using the same dataset:

Tasks

☐ Add **2** to every Age value.

☐ Subtract **1** from every Age value.

☐ Multiply the Area column by **1.1**.

☐ Compute the transpose.

☐ Verify every operation using NumPy.

> **AI Insight**

- Which operation changes values?
- Which operation changes structure?
- Why is transpose useful?

Mission 3 — AI Prediction

Weights

Feature	Weight
Area	2
Bedrooms	20
Age	-5

Tasks

☐ Create the weight matrix.

☐ Perform matrix multiplication.

☐ Compute the prediction score.

☐ Verify using NumPy.

> **AI Insight**

Where is matrix multiplication used in AI?

Mission 4 — Detecting Redundant Features

The company added a new feature.

House	Area	Bedrooms	Age	Rooms
H1	120	3	10	6
H2	150	4	8	8
H3	180	4	6	8
H4	210	5	4	10

Notice:

> **Rooms = Bedrooms × 2**

Tasks

☐ Create the new matrix.

☐ Compute the rank.

☐ Verify using NumPy.

☐ Explain your findings.

> **AI Insight**

Why are redundant features harmful?

Mission 5 — Determinant, Identity Matrix & Inverse

Matrix A

```
| 2   1 |
| 4   2 |
```

Matrix B

```
| 2   1 |
```

```
| 3   2 |
```

Identity Matrix

```
| 1   0 |
| 0   1 |
```

Tasks

☐ Compute the determinant of Matrix A.

☐ Compute the determinant of Matrix B.

☐ Verify both determinants using NumPy.

☐ Verify

```
A × I = A
```

☐ Verify

```
B × I = B
```

☐ Determine which matrix is invertible.

☐ Compute its inverse using NumPy.

☐ Explain why the other matrix has no inverse.

> **AI Insight**

- What does determinant = 0 mean?
- Why can't a singular matrix be inverted?
- Why is the inverse important in Machine Learning?

Mission 6 — Solving Linear Systems

Given

```
2x + y = 7

x − y = 2
```

Tasks

☐ Write the matrix representation.

☐ Solve manually.

☐ Solve using NumPy.

☐ Compare both solutions.

> **AI Insight**

Why are systems of equations important in Machine Learning?

Submit:

✅ Jupyter Notebook

✅ PDF Report